

The Ultimate Rational Meme

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Abstract

The Ultimate Rational Meme is an anti-anti-rational meme that enables the replacement of any non-Ultimate Rational Memes by any other non-Ultimate Rational Memes. 'Anti-anti-rational' means the rejection of anti-rational memes, while non-Ultimate Rational Memes evolve through a Popperian process of conjecture and refutation. An explanation in constructor-theoretic terms is provided. The Enlightenment represents our closest approximation to the Ultimate Rational Meme. In physical terms, this meme crystalizes branches across the Everettian multiverse.

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	Todos/ idea dump:	
	- Postmodernism as against the URM - Frame multiculturalism as an error - Explain more concretely what the Ultimate Rational Memes is: error correction, criticism, openness, tolerance	
	- We all relive Galileo's story when we try to explain the theory of evolution to a hard-core religious zealot - Fractal to explain the Kantian regulative idea - Culture is a meta modulator of human action - Merge meme theory with constructor theory and classical liberalism - Analogous to how a universal computing device can simulate any physical process (Deutsch (1985)[1]), so can the - the UTR doesn't entrench itself for greater fidelity like Nazism would have done, which makes the UTR inherently unstable	

1 Introduction

Dawkins (1976) [2] introduced the concept of *memes* as units of cultural transmission, analogous to genes in biological evolution. Just as genes propagate biological traits through replication, variation, and selection, memes spread cultural information (ideas, behaviors, symbols) through imitation. The success of a meme depends on its ability to replicate and persist in human minds and communication. Memes evolve through a process similar to natural selection, with the fittest memes surviving based on their appeal, simplicity, or utility.

Deutsch (2011) [3] expands upon Dawkins' idea of memes by focusing on how memes influence the evolution of human knowledge and culture. He emphasizes that while memes can replicate

without necessarily being beneficial (e.g., harmful ideologies), true progress in culture stems from *knowledge-generating* memes—those that create explanations and solutions for problems.

Deutsch introduces the idea that knowledge creation is crucial for cultural evolution. Unlike biological evolution, which is largely blind and random, human cultural evolution is purposeful and driven by creativity and problem-solving. He argues that memes evolve when people actively improve upon ideas through conjecture and criticism, a process closely related to Popper’s theory of knowledge growth via falsification.

1.1 Mode of Replication

Memes replicate in one of two ways.

Rational memes replicate by being useful. They invite criticism, promote the growth of knowledge, and evolve through creative thinking and error correction.

Anti-rational memes replicate by disabling the holder’s rational faculties. They resist criticism, suppress knowledge growth, and rely on coercion or authority to maintain their presence in culture.

1.2 The Ultimate Rational Meme

Drawing upon Deutsch’s insights, I conjecture that there exists a universal set of ideas that, when adopted by a culture, are capable of giving birth to any other ideas within the said culture. Having knowledge of these ideas is a jump to universality that enables its knowers to evolve their culture indefinitely. Concretely, this amounts to making it possible to reach any particular physical state that can, in principle, be reached.

I shall refer to this special set of cultural ideas as the *Ultimate Rational Meme* (or equivalently, *Memeplex*).

The universality of the Ultimate Rational Meme is given by the property that it’s not uniquely associated with human beings. Any group of knowledge-creation entities, e.g. aliens or AGIs (artificial general intelligences), would need to adopt the same meme in order for their other rational memes to evolve.

1.3 The Underlying Ideas

Analogous to how truth is conceptualized in Popper’s epistemological framework, I regard the Ultimate Rational Meme as a Kantian regulative idea. It’s an idea that can and should guide our thinking, but not something we may ever know we have reached fully.

Popper’s critical rationalism is our best abstract implementation of The Ultimate Rational Meme.

Note that the Ultimate Rational Meme is more than just critical rationalism. The former can be viewed as an extension built on top of the latter.

Furthermore, an idea has to be spread far and wide to be called a meme. If only an individual in a society adheres to these principles, it’s not a meme.

- Meta layer on top of Critical Rationalism

2 Constructor-Theoretic Lens

Constructor Theory[4] is a new framework for understanding the laws of physics, aiming to describe all physical phenomena in terms of what is possible and what is impossible. Instead

of focusing on laws of motion, Constructor Theory emphasizes the role of *constructors*—entities that cause transformations and make things happen.

For our purposes, one relevant type of constructor is the *abstract constructor*, of which the most important kind is knowledge. I shall adopt the conception of knowledge provided by Deutsch and Marletto[5]: information which, once it is physically instantiated in a suitable environment, tends to cause itself to remain so: it survives criticism, testing, random noise, and error correction.

Knowledge of the Ultimate Rational Meme constitutes a fundamental abstract constructor, which helps all other abstract constructors perform their tasks more accurately.

3 Evolution of Rational Memes

Friendliness towards other rational memes is a core tenet of the Ultimate Rational Memes. Indeed, the Ultimate Rational Meme enables other ideas to proliferate in an open-ended way.

The Ultimate Rational Meme enables any rational idea to be created in the mind of an individual and replicated in other people's minds.

We can use this to explain a great many historical incidents. Let's look at Galileo.

Galileo's fight with the church can be seen as a clash between the universal rational meme and the anti-rational meme du jour (Catholicism). Thinkers like Galileo had to fight a battle on two fronts: (i) convincing his enlightened peers of the veracity of his discoveries, and (ii) convincing his non-enlightened counterparts that changing one's mind is a good thing.

There may be more fundamental abstract constructors. For instance, the axiomatic system of mathematics. Whether mathematics constitutes a fundamental abstract constructor is left as a creative exercise for the reader.

3.1 Rejection of Bad Ideas

All anti-rational memes are bad ideas, but not all bad ideas are memes.

4 Anti-Anti-Rational

Anti-rational memes are collections of ideas such as religions, taboos, cults of personality, and ultra-conservatism.

Since the ultimate rationale meme allows any other rational meme to evolve, it follows that it is anti-anti-rational. It has to be like this because otherwise the anti-rational memes would not allow other memes to take their place.

Furthermore, note that this isn't to imply that in a society, the Ultimate Rational Meme cannot co-exist with anti-rational memes. They can, and they often co-exist.

The point is that in the domains where the Ultimate Rational Meme operates explicitly, such as scientific discovery, anti-rational memes are rejected. It could be said that the Ultimate Rational Meme is operationally immiscible with anti-rational memes.

5 The Enlightenment

If critical rationalism is our best abstract implementation, the political structures brought about by The Enlightenment are our best real-world application of The Ultimate Rational Meme.

Due to the fecundity of scientific progress, it takes as much effort to describe the last 500 years as it takes to describe the rest of history combined.

5.1 Critical Rationalism

It is worth exploring

5.2 Self-Preservation

Postmodernism.

We need to defend the Enlightenment at all costs. It's our best hope to reach the stars.

Wokeness is an anti-rational meme that is actively resisting the Ultimate Rational Meme.

6 Solutions to Paradoxes

I will use the power of the concept of Ultimate Rational Meme to solve two paradoxes in the field of political philosophy.

6.1 Paradox of Tolerance

Popper (1945)[6] proposed the *paradox of tolerance*. The paradox states that unlimited tolerance, if extended to those who are intolerant, can lead to the destruction of tolerance itself. In other words, if a society is too tolerant, especially towards intolerant ideologies or behaviors, those intolerant elements can undermine and eventually overthrow the very foundation of a tolerant society.

However, through the lens of the Ultimate Rational Meme, this situation is not paradoxical.

Fundamentally, the goal of the *intolerant* is to supplant the Ultimate Rational Meme with an anti-rational one. In the case of World War 2, the particular anti-rational meme was Nazi fascism.

As long as we know that we need the Ultimate Rational Meme to continue evolving our other rational memes, there's nothing paradoxical in fending off attempts to supplant the Ultimate Rational Meme with an anti-rational meme.

6.2 Tradition of Criticism

There appears to be a tension between the principle of error correction and the attempt to institutionalize it.

On the one hand, correcting errors means changing things and ideas. On the other hand, to have error correction part of the culture requires the principle itself to remain entrenched and unchanged.

Popper (1945) [popper'open'2013] hinted at this tension in pages 372-390 (TODO: find pages) of "*The Open Society and Its Enemies*". He ended up concluding that critical rationalism requires a little faith on behalf of its adherents.

The Ultimate Rational Meme challenges that conclusion, as well as the view that there's an inherent tension in a tradition of criticism.

The tension is resolved by noticing that institutionalizing the Ultimate Rational Meme is rational. It leads to unbound progress through the evolution of all other rational memes.

7 Physical Lens

As noted in the introduction, the Ultimate Rationale Meme is universal. It extends beyond human societies. The culture of any intelligent alien civilization far away from Earth contains

the same meme.

This has some peculiar implications for physical reality. Using Everett (1957)[7]’s conception of quantum theory, I note that the Ultimate Rational Meme is resistant to differentiation.

Anti-rational memes create different worlds because they are easy to vary at the core. Many of the details (e.g. names and places) in the Bible can be changed and the Christianity memplex as a whole would remain unscathed.

Conversely, rational memes are harder to vary. Change one term in Schrödinger’s equation and quantum theory falls apart.

The Ultimate Rational Meme enables an acceleration of the evolution of rational memes, and thus, it leads to the formation of *abstract multiversal crystals*. In more concrete terms, this means that there can be two branches in the Everettian multiverse that have split at a long time ago, but which have become ideologically alike because in both branches, the Ultimate Rational Meme has led its adherents to discover the laws of physics (which are the same in all branches).

Thus, converging on the truth has physical implications. Knowledge has the profound capability to transcend decoherence, as it were — at least as far abstract patterns are concerned.

8 Connection to brain plasticity

Psychedelics are meta-perspective modulators. They change the mind in such a way that the mind then becomes more open to changing itself about other things. In personality-theoretical terms, it increases the Openness trait.

Future societies that will routinely use psychedelics will experience creativity on steroids.

- Meditation - Nootropics

9 Ending Remarks

The Ultimate Rational Meme is a driving force towards potentially infinite technological and moral progress.

It cannot be otherwise. We need to continue creating new knowledge to e.g. fend off a potential asteroid collision.

Deutsch’s[3] argues that morality consists of *not destroying the means of error-correction*. Equivalently, I argue that morality consists of *not replacing the Ultimate Rational Meme and our best approximation of it, the philosophical movement known as the Enlightenment*.

References

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